



NUA

The AI-driven Restaurant, Retail & Services OS.

PREPARED FOR

Saumil Bhatt

Platform overview · 14 August 2026
From our very first conversation to today.



What NUA POS is today

NUA POS is a full-stack, AI-driven commerce operating system that has grown out of a traditional restaurant POS into a vertical-aware platform for Restaurants, Retail, and Beauty & Services businesses. It combines a real-time point-of-sale, an autonomous operating layer (NUA Agent), enterprise-grade tenant isolation, and a growing marketplace of connected third-party services — all designed for owner-operators who want a single system that runs their business instead of a stack of tools that require running.

This document summarises every capability shipped from our very first conversation through today's sync, plus the differentiators that put NUA ahead of Square / Toast / Lightspeed for the small-to-mid multi-location operator, and a candid look at what's next.

Point of Sale & Operations

- Full-touch POS Terminal with docked modifier panel, product grid, cart, split payments, and offline-safe menu cache.
- iPad-first UX: hand-tuned touch events, swipe-to-delete on cart, tablet-friendly modifier picker, and short-viewport fixes.
- POS Layout builder — owners customise button layout, background colour, and section order per station.
- Guest bill-splitting via table QR — a party can settle their own seats, apply their own tips, and pay independently.
- Real Stripe checkout on online orders, and native Bitcoin / Lightning / USDC via Coinbase Commerce.
- EFTPOS terminal I/O simulator + real terminal history and connection health.
- Kitchen Display with course lifecycle, docket detail, seat notes, average order time, station filters, and auto-refresh.
- Coursing scheduler — minute-level firing rules for pacing courses across the pass.

Inventory, Menu & Purchasing

- AI Menu Import: upload a photo or PDF menu, review parsed items, edit prices, and commit to catalog.
- Real product variants + barcode support (size / colour / material matrices, per-variant SKU and stock).
- Measured / fractional stock — supports draft beer taps, cocktails, bulk retail.
- Nested modifier groups with instant required-choice validation and dock-panel UI.
- Purchase Orders with editable line items, supplier email dispatch, and receive-screen landed-cost pricing.
- Stock transfers between locations with in-transit ledger.
- Predictive inventory events feeding the rules engine — `inventory.predicted_stockout`.
- Category search matching, per-item bulk edit, Recently Edited Today rail, product-image i18n.



Guests, Loyalty & CRM

- Unified Loyalty Engine — earn / redeem / balance / liability / ledger, refund clawback + redeem restoration.
- Loyalty Passport — a single guest's points aggregate across sibling locations.
- Guest Loyalty Portal at /rewards — passwordless guest identity shared across booking, ordering, and loyalty.
- Segment builder with time-windowed spend rules and recurring campaigns.
- Vouchers: public voucher-check endpoint, guest voucher entry on OrderOnline, POS voucher picker, staff inbox.
- Gift cards with unified audit trail across issuance, redemption, and reversal.
- Live waitlist tracking for guests at /track-waitlist.
- Guest OTP hardening + phone-verified identity for high-trust actions (split-bill pay, waitlist join).

Reservations & Appointments

- Reservations Month Calendar with per-day booking counts, cover totals, and green-dot occupancy indicators.
- Booking Blackouts — pause incoming bookings for a specific day or date range with reason.
- Bookings Marketplace — self-serve partner applications flow.
- Staff-as-resource appointment booking (beauty vertical) — no two clients ever land on the same person at once.
- Deposits, no-show fees, and structured client intake notes on appointments.
- Service catalog with per-service duration, price, deposit, and requirements.

Multi-Business, Finance & Compliance

- Multi-Business tenant isolation — every product, promotion, expense, supplier, refund, transaction, voucher, and wallet ledger is scoped by businessId.
- Backfill utility that stamps businessId onto historical records.
- Sales & revenue reporting scoped per business, with P&L, deposits liability, and budget-vs-actual pages.
- Double-entry finance foundation with drawer sessions, End-of-Day close, and tax-line receipts (GST-inclusive pricing).
- Compliance CSV export — approvals + rules + trust trail as one downloadable file.
- Real audit log with field-level diffs; ghost-discount backdoor removed.
- First-login onboarding wizard: business details > ABN verification > vertical > confirmation.
- Business-vertical-aware nav that adapts sidebar and tabs for Restaurant vs Retail vs Beauty.



NUA Agent, Rules Engine & AI

- NUA Agent — an autonomous operating layer that runs on top of live business data.
- Multi-turn tool calling with graduated trust: proposals flow through shadow > approval-queue > auto-approve based on `nua_trust` score.
- Approval Queues so owners stay in the loop on high-impact actions until the agent earns autonomy.
- Cross-module Rules Engine with predictive events (`inventory.predicted_stockout`, `ops.error_spike`) feeding real actions.
- Ash Trust Ladder — shadow-audit review feed with instant demotion when an agent misbehaves.
- AI Concierge with voice/lookup (real Twilio-backed outbound calls — refuses to dial without credentials rather than faking it).
- AI Menu Import via multimodal vision (Emergent LLM key + pypdf for PDF menus).
- Continuous background health monitoring: `/api/health` surfaces DB latency + all scheduler states.

Security, Identity & Reliability

- Real TOTP 2FA with device trust, backup codes, and enforced login challenge.
- Staff PIN login as the primary tap-to-sign-in flow, with roster gate (staff can only sign in during rostered hours).
- Manager PIN approval flow on high-privilege actions (discounts, voids).
- POS session timeout with graceful reauth.
- Passwordless guest identity shared across booking / ordering / loyalty.
- Password reset with resend cooldown and rate-limiting.
- Structured request logging, error observability, and error visibility panel.
- Backup + restore drill (daily automated backup-restore test — catches a silently-broken backup before you need it).
- Ephemeral-collection retention policies + TTL indexes; financial data explicitly excluded from purge.
- Section-grouped, sub-permission staff catalog (~70 granular permissions across 15 sections).



Marketing, i18n & Integrations

- Order Online + Table Order + CFD + Kiosk all fully i18n with a live language selector.
- Channel Menus with per-channel pause/resume, pause-until, and full scheduled hours (Simple daily / Weekly per-day / Date overrides for holidays).
- NUA Connect: real integration framework + Square connector, honest provider-status ('never faked').
- Live Happy Hour countdown on CFD.
- Recurring segment campaigns.
- What's New in-app changelog feed so owners see shipped features without leaving the app.
- Route-based code splitting — first-load bundle roughly 70% smaller.
- Real-time push notifications and offline queue for POS actions.

What makes NUA stand out in today's market

Compared to Square, Toast, Lightspeed and the broader 'app-store POS' incumbents — where you rent 12 apps and manually stitch them together — NUA ships these as a single coherent product:

- Vertical-aware from day one (Restaurant / Retail / Beauty) — one platform, not three separate products.
- AI as an *operating layer*, not a chatbot bolt-on: NUA Agent proposes real actions, earns trust via a graduated ladder, and runs 24/7 on live data.
- Continuous auto-sync from GitHub main to preview at 4am AEST — the app self-updates without redeployment friction.
- Multi-tenant + Loyalty Passport out of the box — points and guests follow the customer across sibling locations, not per-location silos.
- Honest integration status — no faked 'connected' badges. If Twilio credentials aren't set, the voice endpoint clearly refuses to dial.
- Native crypto acceptance (BTC / Lightning / USDC) alongside Stripe, on the same 'redirect, webhook confirms' sale-finalize pipeline.
- Guest bill-splitting via table QR — party members pay independently without the server retyping anything.
- Autonomous background jobs (Ash scheduler, Coursing scheduler, Backup drill, Repo-sync) that keep the app improving on its own.
- TOTP 2FA + PIN + roster gate + manager approval — enterprise-grade access control, not just a password.
- Every merge shipped with tests. In-process pytest suite gates every deploy — no 'move fast and break payments'.



Upcoming features & potential enhancements

What I'd prioritise next — mix of user-requested backlog and net-new ideas that would compound on what's already shipped.

P0 — user-verify & polish (short backlog)

- Push preview > GitHub via Save to GitHub so remote catches up with the credential-alignment fix and v36.6 work.
- Production ADMIN_EMAIL / ADMIN_PASSWORD env alignment on the deploy panel — trigger seed_admin to reset the owner password on next boot.
- Full sweep of the newly-shipped verticals: onboarding wizard on a fresh account, ABN verification path, per-vertical nav switching.

P1 — high-impact enhancements

- Real Meta / TikTok / X OAuth for social scheduling — currently mocked; wiring real OAuth would unlock the marketing side.
- SendGrid + Twilio production credentials to activate the notification abstraction (currently best-effort).
- Hardware Health polish — printer, cash drawer, EFTPOS device status feeding a real dashboard.
- 'Require 2FA for owner/manager voids and refunds' — small toggle, big compliance win for hospitality audits.
- QR-on-receipt-PDF that opens the guest's own loyalty balance — zero-friction repeat-visit lever.
- Punctuality trend mini-chart per staff — catch 'trending late' weeks before it becomes a real problem.
- 'Reopens Mon 11am' auto-notification to guests trying to order from a paused channel.
- Stripe test-mode banner in Settings so staff know when they're on test keys.

P2 — differentiator ideas

- Keyboard shortcut cheatsheet overlay (press ? anywhere) — Back, quick-nav, bulk actions. iPad-with-keyboard operators will love it.
- Modifier panel remembers last modifier group per product per regular — two-tap ordering for the morning rush.
- Table share to guest phone via QR — extends bill-split into a full self-serve pay flow.
- Slack webhook when the daily 4am-AEST repo-sync lands new commits — 'What's New' pushed to your phone.
- Sync History tile on the Today dashboard for at-a-glance visibility.
- 'Set PIN' button in the manager staff-management page for one-tap PIN issuance to new hires.
- @dnd-kit tablet-friendly drag-and-drop for the Social Calendar.
- x-ai-parsed-fallback header on AI endpoints so the client knows when a response came from a fallback path.
- Refresh the broken Unsplash product-image URLs to a local placeholder set.
- Move the ai_weekly_plan generation into a background task rather than a synchronous request.

Longer-horizon (Phase 2/3 from the roadmap)

- Full AI phone/voice channel that answers inbound bookings and orders end-to-end.
- Consolidate the four fragmented loyalty subsystems per the 763-line Loyalty Engine Spec (docs/LOYALTY_ENGINE_SPEC.md).
- Enterprise-grade double-entry Finance module: ledger, journals, reconciliation, and audited



close.

- Highly advanced inventory mechanics — measured/fractional stock extended into recipe-level yield tracking.

One last thing.

Everything in this document was built collaboratively — from the first cart-swipe bug to the vertical-aware onboarding wizard, from Channel Schedule to the daily auto-sync. It's your platform, grown one merge at a time.